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S^{3}DNet: Support-Driven Spatial–Semantic Decoupling for Few-Shot Strip-Steel Surface Defect Segmentation

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Abstract

Few-shot strip-steel surface defect segmentation requires a model to identify unseen defect categories and delineate their precise boundaries from only a few annotated support images. This remains challenging because strip-steel defects are often small, elongated, low-contrast, and easily confused with repetitive background textures. To address these challenges, we propose the Support-Driven Spatial–Semantic Decoupling Network (S^{3}DNet), which preserves a frozen DINOv3 encoder while explicitly separating support-conditioned semantic adaptation from fine-grained spatial refinement. Specifically, a low-rank prototype adapter transforms the masked support prototype into an episode-conditioned semantic update of the query tokens, enabling the frozen representation to adapt to the defect category specified by the current support set. In parallel, a lightweight spatial adapter extracts multiscale local cues for recovering weak boundaries and thin structures. A spatial–semantic interaction module subsequently uses support–query semantic similarity to selectively gate these spatial cues before progressive decoding, thereby suppressing irrelevant texture responses while retaining defect-relevant details. Experiments on FSSD-12 demonstrate that S^{3}DNet consistently outperforms representative CNN-based few-shot segmentation methods under both 1-shot and 5-shot settings, while the lightweight DINOv3-Small configuration achieves a favorable accuracy–efficiency trade-off. Ablation and qualitative analyses further validate the effectiveness and complementarity of the proposed semantic adaptation, spatial refinement, and interaction mechanisms.

Keywords: few-shot semantic segmentation; industrial defect inspection; strip-steel surface defects; vision foundation models; parameter-efficient adaptation; support-conditioned learning

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1. Introduction

Strip-steel surface inspection is a fundamental component of industrial quality control. A reliable inspection system must not only identify whether a defect exists, but also delineate its spatial extent and boundary so that the result can support severity assessment, product grading, and process diagnosis. Modern segmentation networks have achieved promising performance when large pixel-wise annotated datasets are available. However, collecting such annotations in a production line is expensive and time-consuming. Rare

fault types may appear only a few times, and the appearance distribution can change with illumination, material texture, rolling speed, and imaging configuration. Consequently, a model trained with abundant samples from a fixed set of categories is difficult to maintain and rapidly redeploy when a new defect type occurs [1].

Few-shot semantic segmentation [2–4] provides a practical alternative. Instead of retraining a model with many examples, it specifies a target category through a small support set and predicts the corresponding mask in a query image. This setting is naturally compatible with industrial inspection because engineers can introduce a new defect by annotating only one or several representative images. Nevertheless, strip-steel defects are substantially more difficult than many objects in natural scenes. They often occupy only a small proportion of the image, appear as narrow scratches or fragmented spots, and have weak intensity differences from the background. Meanwhile, repetitive steel texture can resemble the target pattern. The model must therefore simultaneously infer a class-consistent representation from very limited support evidence while retaining sufficient spatial resolution to recover subtle boundaries [5,6].

Existing few-shot segmentation methods predominantly build prototypes, prior masks, or dense correlations from features extracted by a convolutional backbone. Convolutional representations preserve local continuity and hierarchical spatial detail, which are useful for boundary localization. However, under limited annotation and large category shifts, their semantic responses can become unstable and may activate strongly on background edges that resemble the support texture. Vision Transformers provide a complementary property. Their global self-attention facilitates long-range aggregation and produces more coherent semantic representations, but patchification and repeated feature abstraction may smooth high-frequency details. For tiny or low-contrast defects, a semantically correct response can consequently spread beyond the boundary or fail to cover a thin structure. Directly replacing a CNN with a Transformer therefore does not by itself resolve the conflict between semantic reliability and spatial precision.

Self-supervised vision foundation models provide a stronger representation starting point. DINO learns semantically organized features and object-aware attention without manual labels [7]. DINOv2 further improves the robustness and transferability of frozen visual representations through large-scale curated pretraining [8]. More recently, DINOv3 addresses the degradation of dense features during long training through Gram anchoring, making its intermediate representations particularly attractive for downstream dense prediction [9]. These developments suggest a parameter-efficient transfer paradigm: the foundation encoder is preserved, while a limited set of task-specific parameters supplies the information required by the new industrial task. Such a strategy is desirable when annotation is scarce, the deployment cycle is short, and full backbone fine-tuning would introduce overfitting or excessive computation.

The value of frozen DINO features has been demonstrated in industrial anomaly detection. MVTec AD provides a representative benchmark containing diverse real-world anomalies [10], and methods such as AnomalyDINO exploit DINOv2 patch features to obtain competitive localization with few training samples [11]. Foundation encoders have also been studied as few-shot anomaly detectors with little or no task-specific backbone adaptation [12]. However, anomaly detection and the task considered in this paper are fundamentally different. The former generally learns a distribution of normal samples and identifies deviations, whereas few-shot defect segmentation receives an explicitly masked support set and must distinguish one target class from both the normal background and other possible defect patterns. The support information must therefore influence the semantic representation, rather than serving only as a reference bank after feature extraction. For this reason, we do not treat anomaly detectors as quantitative baselines in this work:

their protocol contains no annotated support mask, no episodic unseen-class split, and no class-specific mask output, so a direct numerical comparison on FSSD-12 would measure the difference between two task definitions rather than the difference between two methods under a shared protocol.

When a vision foundation model is transferred to this setting, two challenges become particularly important. **(i) Support-conditioned semantic adaptation.** A frozen encoder provides general semantics but does not know which unseen defect is specified in the current episode. Naively fine-tuning the entire encoder is parameter-intensive and can destroy the transferable representation. A compact mechanism is needed to inject the support mask and prototype into the query features. **(ii) Target-aware spatial refinement.** Dense DINO tokens are semantically strong, yet a patch grid alone may be insufficient to capture elongated, weak, or very small defects. Introducing an independent convolutional path can recover texture and edge information, but an unguided path may amplify the repetitive background. The spatial evidence must therefore be filtered by the current support semantics before fusion [13,14].

Figure 1 gives an intuitive comparison. The ResNet-50 response emphasizes local edges and captures part of the defect structure, but it is fragmented and includes unrelated background locations. DINOv3 forms a more globally consistent response, yet the boundary is smoother and several regions remain insufficiently localized. In contrast, our model preserves a coherent target response while suppressing scattered activation and recovering the principal geometry. This observation motivates an explicit decoupling design instead of forcing a single feature path to satisfy two conflicting requirements.

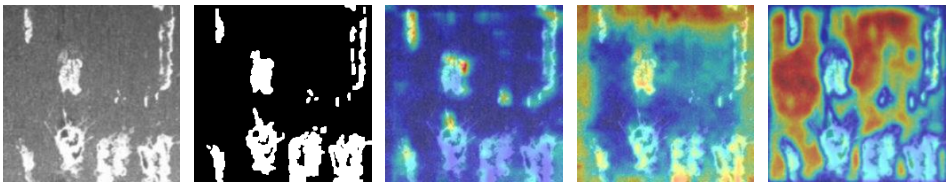


Figure 1. Spatial responses produced by different representation mechanisms for the same defect sample. ResNet-50 captures local texture but yields fragmented activation; DINOv3 is semantically coherent but spatially smooth; the proposed model concentrates its response on the defect while retaining its structure.

To this end, we propose the Support-Driven Spatial–Semantic Decoupling Network (S^3DNet). A frozen DINOv3 encoder acts as the semantic foundation. At selected intermediate layers, a low-rank prototype adapter (LPA) aggregates support tokens with the support mask and maps the resulting prototype through a low-rank bottleneck. The compact update conditions the query tokens on the target class without changing the backbone parameters. In parallel, a low-weight spatial adapter (LSA) extracts three scales of local structure using shallow convolutions. A spatial–semantic interaction module (SSI) then computes support–query cosine similarity, converts it into a spatial gate, and uses the gate to select defect-relevant LSA features. Finally, a top-down decoder progressively integrates high-level semantics with high-resolution structure and produces the query mask. In this way, the semantic branch determines what should be segmented, while the spatial branch helps determine where its fine boundary lies.

The main contributions of this work can be summarized as follows:

- We develop a support-driven spatial–semantic decoupling framework for few-shot strip-steel defect segmentation. It preserves a self-supervised DINOv3 encoder and introduces only compact adaptation, spatial, interaction, and decoding components, providing a practical transfer strategy for data-scarce industrial inspection.

- We propose three complementary modules. LPA performs low-rank support-conditioned semantic adaptation, LSA restores multiscale local structure with small overhead, and SSI transforms prototype similarity into target-aware spatial gating for aligned progressive fusion.
- We conduct three-fold 1-shot and 5-shot evaluation on FSSD-12, including comparisons with representative methods, two foundation-model configurations, computational analysis, component ablations, qualitative results, and failure-case discussion. The results consistently validate both the segmentation accuracy and the accuracy–efficiency trade-off of S^3DNet.

2. Related Work

2.1. Industrial Surface Defect Segmentation

Surface defect segmentation has traditionally been developed from fully supervised dense prediction architectures. Fully convolutional networks [15] first replaced image-level classifiers with end-to-end spatial prediction, U-Net [16] combined an encoder–decoder structure with skip connections to recover fine details, and DeepLabv3+ [17] integrated atrous context modeling with a boundary-aware decoder. These designs provide the basic multiscale mechanisms used in industrial inspection, but their success generally depends on abundant pixel-level annotations. Such supervision is expensive for production lines because defect occurrence is sparse, acquisition conditions change over time, and newly emerging defect types must be labeled by experienced inspectors.

Strip-steel inspection introduces additional difficulties, including low contrast, strong rolling texture, large scale variation, and elongated or fragmented targets. Wen et al. [1] reviewed the progress and persistent challenges of steel-surface inspection. For hot-rolled strip steel, Feng et al. [18] combined ResNet50 with channel and frequency-aware attention to improve defect classification. More recently, SFF-Net [19] focused convolution and proposal generation on supervised defect regions, reducing redundant computation over large uniform backgrounds. These detection and classification methods improve localization or recognition under a fixed label set, but normally require additional training data when the inspection target changes.

Industrial anomaly detection and uncertainty-aware segmentation reduce annotation demands or improve prediction reliability from a different perspective. MVTec AD [10] established a widely used benchmark for anomaly localization. Neural subspace learning [20] integrates low-rank and sparse priors into an autoencoder and explicitly addresses the small-sample problem in surface inspection. Conformal segmentation [21] instead calibrates pixel prediction sets to provide statistical control of expected detection errors. Although these approaches are valuable when normal data or calibration data are available, anomaly detection and few-shot semantic segmentation have different objectives: the latter must isolate the foreground class specified by an annotated support example, even when other abnormal structures are present.

Few-shot industrial segmentation therefore treats the annotated defect itself as the task specification. TGRNet [5] constructs triplet graphs between support, query-foreground, and query-background representations, and CPANet [6] aggregates cross-position evidence to cope with appearance variation in strip-steel defects. Selective Prototype Network [22] suppresses ambiguous support information for metal-surface segmentation. Beyond the few-shot setting, MSOFormer [23] demonstrates that mask-space optimization and spatially informed Transformer queries can improve segmentation of small, irregular industrial defects under severe foreground–background imbalance. Recent few-shot studies combine metric learning with foundation-model representations [13] or emphasize small-target transfer in visual inspection [14]. Very recently, P^3-SAM [24] adapted SAM to few-shot

strip-steel defect segmentation through perceptual parallel prompting, confirming that promptable foundation models require industrial-specific adaptation when defects exhibit low contrast and fine-grained texture. Together, these developments show that industrial few-shot segmentation requires both transferable category evidence and accurate recovery of weak, spatially sparse structures.

2.2. Few-Shot Semantic Segmentation

Few-shot semantic segmentation (FSS) predicts a query mask from one or several labeled support images whose categories are absent from training. Early work [2] formulated one-shot segmentation as conditional dense prediction and subsequently introduced metric learning to improve category transfer. Prototypical networks [3] represent a class by the mean of its embedded examples. SG-One [4] uses similarity guidance to condition query segmentation, and PANet [25] adds prototype alignment so that support and query predictions regularize each other. CANet [26] instead refines a class-agnostic prediction through iterative support–query interaction. FSS-1000 [27] broadens category diversity and facilitates evaluation beyond the common natural-image splits.

A single averaged prototype is compact but may under-represent categories with diverse appearance or disconnected parts. Prototype mixture models [28] introduce multiple semantic components instead of a single center, and adaptive prototype learning [29] dynamically allocates prototypes according to object complexity. Selective and context-aware prototype construction has likewise been used to reduce foreground contamination in industrial scenes [22]. These methods improve intra-class coverage, although prototype quality remains sensitive to the resolution and semantic stability of the underlying feature maps.

Another line of research strengthens dense support–query correspondence. PFENet [30] derives a training-free prior mask from high-level similarity and enriches query features with multiscale context; PFENet++ [31] further filters noisy context to make this prior more reliable. HSNet [32] constructs a four-dimensional hypercorrelation and progressively squeezes it into a query representation, while Democratic Attention Networks [33] distribute matching evidence over multiple correspondences. SCL [34] combines self-guided and cross-guided signals to improve intra-object consistency. PGNet [35] connects distant query regions through graph-based reasoning, DCP [36] decomposes a difficult category into proxy subproblems, SCCAN [37] calibrates cross-attention between support and query features, and DRNet [38] disentangles then recombines transferable components. Dense interaction provides expressive matching, but its computational cost and reliability depend strongly on whether support foreground, query target, and structured background are already well separated in feature space. Recently, training-free pipelines have become a visible trend: Sun et al. [39] achieved cross-domain FSS without episodic training by matching robust semantic representations of foundation models, observing that episodic training can cause overfitting even when powerful encoders are incorporated.

2.3. Vision Foundation Models for Dense Prediction

Transformers have reshaped visual representation learning by modeling long-range interactions over image patches. ViT [40] demonstrates that a pure Transformer can learn competitive visual representations at scale, while Swin Transformer [41] introduces shifted local windows and a hierarchical feature pyramid for dense tasks. Self-supervised learning further reduces dependence on manual category labels. DINO [7] reveals emergent object-level organization in attention and patch tokens, while MAE [42] learns scalable representations by reconstructing missing patches. DINOv2 [8] improves transfer through large-scale curated pretraining, and DINOv3 [9] uses Gram anchoring to maintain the qual-

ity and spatial consistency of dense features during long training. These models provide semantically stable patch representations that can be reused without fitting a full encoder to a small downstream dataset.

Foundation models have also expanded through promptable supervision. The Segment Anything Model (SAM) [43] learns prompt-conditioned masks from large-scale segmentation data. Nevertheless, promptable mask generation and episodic FSS are not identical. FSS requires the support image and its pixel mask to define a previously unseen class, and industrial targets may be substantially smaller or less object-like than the regions emphasized during natural-image pretraining. Recent efforts nevertheless narrow this gap: Tsai et al. [44] revisited FSS from the perspective of promptable foundation models and achieved competitive training-free segmentation with SAM3.

In industrial vision, frozen foundation features have shown strong anomaly-localization ability. AnomalyDINO [11] applies DINOv2 patch features to few-shot anomaly detection without updating the encoder, and related analysis interprets pretrained visual encoders as implicit few-shot anomaly detectors [12]. These results motivate the design of S^{3}DNet: large-scale pretraining supplies robust semantics, but patch tokens alone may not retain the weak boundaries and high-frequency detail needed for thin defects. Consequently, adapting foundation features to industrial segmentation involves both target-specific semantic conditioning and explicit spatial refinement. For FSS itself, FSSDINO [45] showed that class-specific prototypes built on frozen DINOv3 features are already competitive with specialized FSS decoders without any training, supporting the premise of this work that frozen self-supervised features carry transferable category evidence. Accordingly, MVTec AD and AnomalyDINO are discussed here as motivation, evidence that frozen foundation features transfer to industrial data. But they are excluded from the quantitative comparison in Section 4: MVTec AD is built on normal-only training splits and evaluates threshold-based anomaly localization, whereas episodic FSS predicts class-specific masks for support-specified unseen categories, and the two protocols do not share a comparable metric.

2.4. Parameter-Efficient Adaptation and Spatial Refinement

Parameter-efficient transfer learning preserves most pretrained weights and updates only a small task-specific component. Houlsby adapters [46] insert compact bottlenecks into a frozen network, and LoRA [47] represents weight updates by low-rank factors. In vision, Visual Prompt Tuning [48] learns a small set of input tokens, AdaptFormer [49] adds lightweight parallel branches to Transformer blocks, and SSF [50] provides a particularly compact tuning baseline through feature scaling and shifting. Their common objective is to retain broad pretrained knowledge while reducing storage, computation, and overfitting for each downstream task. Most such approaches learn a single task-level modification; episodic segmentation additionally benefits from conditioning the modification on the current support class.

Accurate masks also require mechanisms that recover fine spatial detail. Feature Pyramid Networks [51] propagate high-level semantics through a top-down multiscale pathway, and Segmenter [52] reshapes Transformer tokens into dense predictions. A Transformer encoder–decoder with token spatial information fusion [53] further shows that global self-attention benefits from explicit convolutional local context in semantic segmentation. However, tokenization and repeated abstraction can weaken local gradients. LaplacianFormer [54] explicitly introduces frequency-sensitive modeling to strengthen multiscale boundaries. Convolutional detail branches are an efficient alternative because their locality is well suited to texture transitions, but unconditional fusion can amplify background

edges as easily as true defect boundaries. A reliable industrial decoder therefore requires semantic evidence to gate which spatial cues are fused.

Summary and positioning. Existing FSS methods offer effective prototype or correlation mechanisms, foundation encoders provide transferable semantics, and lightweight spatial pyramids preserve local structure; nevertheless, treating these ingredients independently does not resolve the conflict between semantic generality and boundary sensitivity in few-shot industrial scenes. In contrast, S^{3}DNet delegates explicit and complementary roles to these components. A support-driven low-rank adapter conditions frozen DINOv3 semantics on the current defect, a shallow convolutional adapter supplies class-agnostic spatial candidates, and semantic similarity gates those candidates before progressive decoding. Compared with training-free foundation-model pipelines, which match frozen features without task-specific learning, and with promptable models, which depend on test-time prompts, the learned low-rank update writes the support semantics directly into the query representation once per episode and requires no test-time prompting. This spatial–semantic decoupling is designed to retain the transferability of the foundation representation while preventing repetitive background texture from dominating fine-detail recovery.

3. Method

3.1. Overview

Few-shot defect segmentation must answer two different questions from limited supervision: which visual pattern corresponds to the support defect, and where its precise boundary lies in the query image. A single feature pathway is not ideally suited to both. Highly pretrained Transformer features are transferable and semantically stable, but their patch-level representation may smooth narrow scratches and weak edges. Shallow convolutional features retain such local evidence, yet they also respond strongly to irrelevant rolling texture, illumination changes, and sensor noise. The main idea of S^{3}DNet is therefore to separate semantic adaptation from spatial refinement and allow the former to gate the latter.

For a K-shot episode, the support set is denoted by $\mathcal{S} = \{(I_s^{(k)}, Y_s^{(k)})\}_{k=1}^{K}$, where $I_s^{(k)}$ and $Y_s^{(k)}$ are a support image and its binary foreground mask. Given a query image I_q, the model predicts

$$\hat{Y}_q = f_\theta(I_q, \mathcal{S}), \tag{1}$$

where the test categories are disjoint from the training categories. The entire DINOv3 encoder is frozen, including its patch-embedding layer, all Transformer blocks, and all layer-normalization parameters, so that no within-backbone adaptation such as LayerNorm tuning is involved, whereas θ contains only the lightweight semantic adapters, spatial branch, interaction modules, and decoder. This choice preserves the general representation acquired during large-scale self-supervised pretraining and restricts task-specific learning to components that use episodic support evidence.

As illustrated in Figure 2, the proposed architecture consists of four functional stages. First, the frozen encoder extracts multilevel support and query tokens. The low-rank prototype adapter (LPA) summarizes the masked support foreground and converts it into a compact update of query semantics. Second, the low-weight spatial adapter (LSA) constructs a shallow feature pyramid directly from the query image, supplying boundary and texture candidates that are complementary to Transformer tokens. Third, the spatial–semantic interaction module (SSI) measures support–query semantic agreement and uses it to gate the spatial candidates at each scale. Finally, a progressive decoder propagates the selected information from coarse semantic localization to fine mask recovery. In this way, support information specifies the target in the semantic path and simultaneously determines where local spatial evidence is trustworthy.

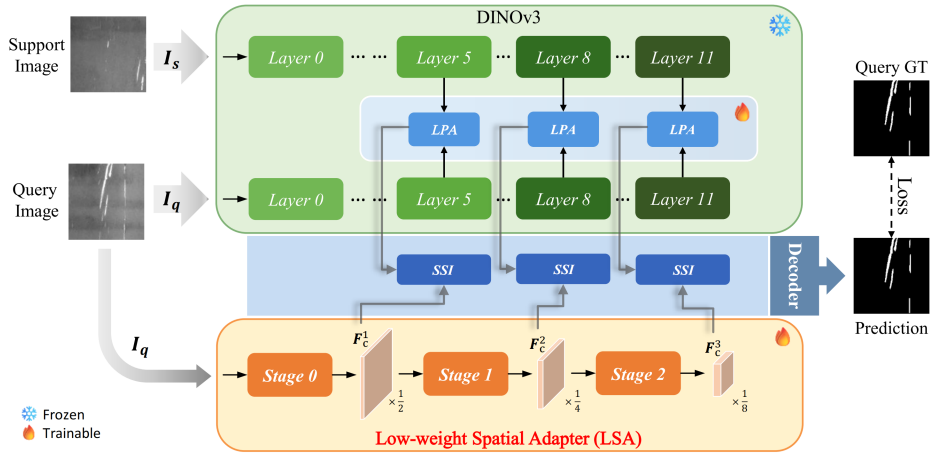


Figure 2. Overall architecture of S^{3}DNet. A frozen DINOv3 encoder provides transferable multilevel tokens. LPA performs support-conditioned semantic adaptation, LSA extracts class-agnostic multiscale spatial cues, and SSI fuses those cues according to support–query similarity. A progressive decoder produces the final query mask.

Concretely, the data flow from input to output proceeds as follows. (i) The frozen DINOv3 encoder maps the support and query images into multilevel patch tokens. (ii) At each selected depth, the masked support tokens are averaged into a prototype, which LPA converts through a low-rank bottleneck into an additive update of the query tokens. (iii) In parallel, LSA convolves the original query image into a three-scale spatial pyramid. (iv) At each scale, SSI compares the adapted query semantics with the scale-specific support prototype and uses the resulting similarity gate to filter the spatial features, which are then concatenated with the semantic maps. (v) Finally, the progressive decoder fuses the three scales from coarse to fine and outputs the binary query mask at the original resolution.

3.2. Support-Conditioned Semantic Adaptation

3.2.1. Multilevel Foundation Representation

The semantic path is responsible for recognizing an unseen defect from one or a few labeled examples. DINOv3 is adopted because its self-supervised patch features retain strong object organization and transfer across categories, while dense-feature regularization improves their suitability for spatial prediction [9]. The same frozen encoder processes support and query images, ensuring that their tokens lie in a shared metric space. Intermediate outputs are sampled from three representative depths and projected to a common channel dimension. Each level passes through a single bias-free linear projection that maps the token dimension of the frozen encoder ($D = 384$ for ViT-S/16 and $D = 768$ for ViT-B/16) to $d = 256$; the three projections are trainable and shared between the support and query branches. The deeper output provides category-consistent context, whereas the earlier outputs retain progressively finer appearance information. Specifically, layers 5, 8, and 11 (zero-based indices) are chosen from the 12-block encoder so that the sampled depths span its shallow, middle, and final stages at a uniform spacing of three blocks; the deepest output therefore coincides with the last Transformer block and yields the most category-consistent semantics, while the shallower ones retain finer detail, and the uniform spacing avoids aggregating adjacent blocks whose representations are highly correlated. After reshaping and interpolation, the three semantic maps are aligned with the coarse, middle, and fine resolutions of the spatial branch.

The input to the Transformer is enlarged before tokenization so that a small defect occupies more patch locations; the empirical effect of this choice is quantified in Section 4.4. This operation does not introduce a trainable super-resolution module; its purpose is simply

to expose a denser semantic grid to subsequent matching. Selecting only three intermediate levels also avoids redundant aggregation of every Transformer block. Hence, the semantic hierarchy provides sufficient scale diversity while retaining the efficiency and stability of a frozen backbone.

3.2.2. Low-Rank Prototype Adapter

Although the frozen representation is broadly transferable, it does not know which foreground category is requested in the current episode. As shown in Figure 3, LPA addresses this problem by turning the annotated support region into an episode-specific semantic direction. At encoder level l, let $S_l, Q_l \in \mathbb{R}^{T\times D}$ denote the support and query patch tokens, and let $m_l \in \{0,1\}^T$ be the support mask resized to the token grid. Masked average pooling yields

$$p_l^{(k)} = \frac{\sum_{t=1}^{T} S_{l,t}^{(k)} m_{l,t}^{(k)}}{\sum_{t=1}^{T} m_{l,t}^{(k)} + \epsilon}, \qquad \bar{p}_l = \frac{1}{K}\sum_{k=1}^{K} p_l^{(k)}. \tag{2}$$

The mask removes most background contamination, and the K-shot average reduces dependence on an atypical support instance. This follows the central principle of prototypical learning [3], but the prototype is used to adapt the representation rather than only serving as a final classifier.

The support prototype is passed through a low-rank bottleneck,

$$\Delta q_l = \alpha_l B_l \sigma(A_l \bar{p}_l), \tag{3}$$

where $A_l \in \mathbb{R}^{r\times D}$ and $B_l \in \mathbb{R}^{D\times r}$ are learnable projections, $r \ll D$, $\sigma(\cdot)$ is GELU, and α_l controls the update magnitude. The adapted query tokens are

$$\widetilde{Q}_{l,t} = Q_{l,t} + \Delta q_l, \qquad t = 1, \ldots, T. \tag{4}$$

The low-rank structure limits the number of trainable parameters and constrains the update to a compact task-relevant subspace. Because the same support-derived vector is added to all query tokens, LPA changes their channel-wise semantic preference without erasing the spatial variation already encoded by DINOv3. Both projections are implemented as bias-free linear layers; A_l is initialized with a Gaussian of standard deviation 0.02, whereas B_l is zero-initialized and α_l starts from one, so that the initial update Δq_l is exactly zero and the network starts from the frozen representation, learning the support-dependent correction progressively.

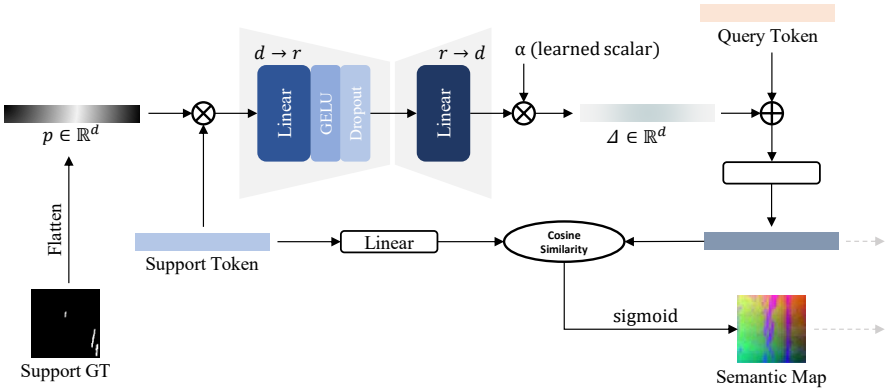


Figure 3. Low-rank prototype adapter. The foreground support tokens are summarized into a prototype, transformed through a compact bottleneck, and added to the query tokens to emphasize the semantic direction specified by the current episode.

LPA differs functionally from task-level parameter-efficient tuning. A conventional adapter learns one fixed modification for an entire downstream dataset, whereas the update in (3) changes with the annotated support set. The same frozen encoder can thus be redirected toward different unseen defects without per-class fine-tuning. Moreover, prototype compression avoids constructing a dense support–query correlation volume, making the semantic conditioning economical for high-resolution industrial images. The resulting maps provide the category evidence used both by the decoder and by SSI to select spatial details.

3.3. Lightweight Spatial Prior Extraction

The semantic path can localize a defect consistently, but its patch grid is not sufficient for every boundary. Thin scratches, tiny inclusions, and weak contrast transitions may disappear as tokens become more abstract. LSA is introduced to expose these local patterns before they are lost. It processes the original query image with three lightweight convolutional stages and produces

$$\{F_c^1, F_c^2, F_c^3\} = \mathrm{LSA}(I_q), \tag{5}$$

where the stages correspond to fine, middle, and coarse spatial resolutions. Each stage contains exactly one convolutional layer, namely, a bias-free 3×3 convolution with stride two and padding one, followed by batch normalization and GELU activation without additional pooling; starting from the input image, the three stages output 16, 32, and 64 channels at $1/2$, $1/4$, and $1/8$ of the input resolution, respectively, so that for a 224×224 input the pyramid resolutions are 112×112, 56×56, and 28×28, matching the aligned semantic maps. The fine feature retains edges and narrow structures, the middle feature describes short-range texture organization, and the coarse feature supplies a spatial layout compatible with high-level semantics.

LSA is intentionally shallow. Its role is not to relearn defect categories from the small training set, since doing so would duplicate the foundation encoder and increase overfitting. Instead, it acts as a class-agnostic prior generator: convolutions identify candidate discontinuities, connected patterns, and local texture changes regardless of their category. This restricted responsibility keeps the branch efficient and makes its complementarity to LPA explicit.

Not every response from LSA should enter the prediction. Strip-steel backgrounds contain repeated edges and high-frequency patterns that can be stronger than the actual defect. Direct addition or concatenation would therefore mix useful boundary evidence with irrelevant structure. The spatial features remain uncommitted until SSI evaluates their semantic consistency. This delayed fusion is crucial: LSA supplies where fine detail may exist, while the support-conditioned semantic path determines whether that detail belongs to the requested defect.

3.4. Support-Driven Spatial–Semantic Interaction

As shown in Figure 4, SSI connects the two decoupled paths. Its purpose is not merely to merge features, but to convert semantic agreement with the support foreground into a spatial selection rule. At scale l, let $F_s^l, F_q^l \in \mathbb{R}^{C \times H_l \times W_l}$ be the projected support and adapted query semantic maps, and let M_l be the support mask at the same resolution. The projection is the bias-free linear layer introduced in Section 3.2.1: applied token-wise, it is equivalent to a 1×1 convolution and maps the encoder token dimension ($D = 384$ for ViT-S/16 and $D = 768$ for ViT-B/16) to $C = 256$, identically for the support and query branches. The scale-specific support prototype is

$$P_l = \frac{\sum_{u,v} F_s^l(u,v) M_l(u,v)}{\sum_{u,v} M_l(u,v) + \epsilon}. \tag{6}$$

Unlike the token-space prototype in LPA, P_l is computed after semantic projection and spatial alignment. It therefore describes the target at the exact representation level used for fusion.

Each query location is compared with P_l using normalized cosine similarity:

$$A_l(u,v) = \text{sigmoid}\left(\tau_l \frac{\langle F_q^l(u,v), P_l \rangle}{\|F_q^l(u,v)\|_2 \|P_l\|_2 + \epsilon}\right), \tag{7}$$

where τ_l is a learnable scale controlling the selectivity of the response. The resulting map A_l indicates how strongly each query position agrees with the foreground support semantics. It then gates the corresponding spatial feature,

$$\widetilde{F}_c^l = F_c^l \odot A_l. \tag{8}$$

Thus, a local edge is retained when it occurs at a semantically plausible target location and attenuated when it is associated with unrelated background texture. Coarser gates encourage global consistency, whereas finer gates preserve target-aligned boundaries and small components.

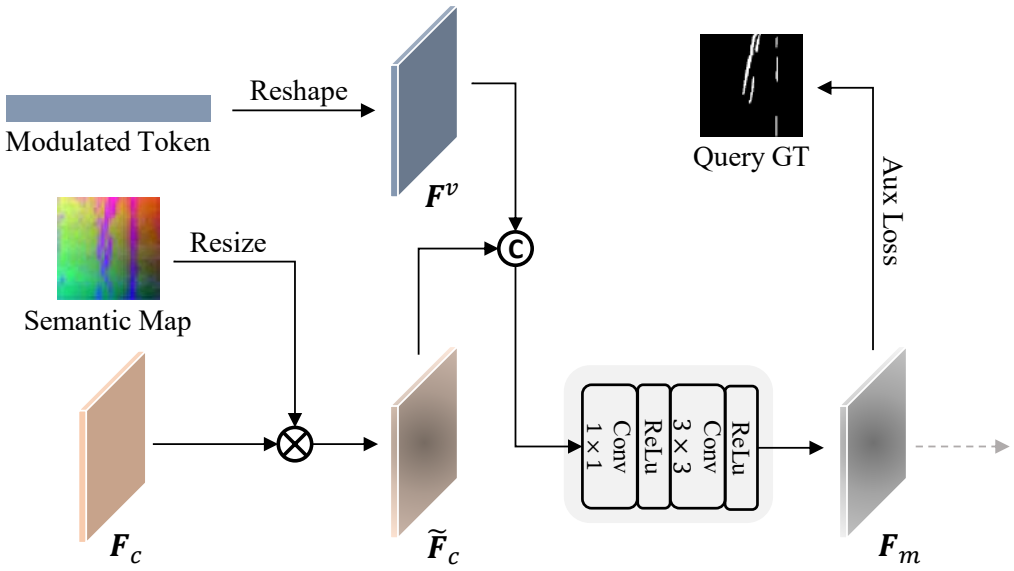


Figure 4. Spatial–semantic interaction module. Support–query semantic similarity forms a spatial gate, which filters the local feature before it is fused with the adapted semantic representation.

The gated spatial feature is channel-wise concatenated with F_q^l and fused through a compact bottleneck: a 1×1 convolution reduces the concatenated channels to 256, followed by ReLU activation, spatial dropout ($p = 0.2$), and a 3×3 convolution that restores 256 channels. Since the semantic map participates in both gate construction and final fusion, it remains the principal source of category identity; the spatial branch supplies a controlled correction rather than an independent prediction. In this way, SSI resolves the main tension of the architecture: high-frequency information is restored where needed without allowing class-agnostic texture responses to override transferable semantics.

3.5. Progressive Decoding and Objective

After interaction at all three scales, the fused representations are decoded from coarse to fine. The coarse map first establishes a stable target region from the strongest semantic context. It is upsampled and combined with the middle-scale map to refine the object layout, and the result is subsequently merged with the finest map to recover weak edges and narrow structures. This top-down propagation follows the general multiscale principle of feature pyramids [51], but differs in that every lateral spatial feature has already been selected by support-conditioned semantic evidence. A final classifier restores the query resolution and produces foreground and background logits. Each merge point bilinearly upsamples the coarser map by a factor of two and concatenates it with the finer map, and a 3×3 convolution then compresses the resulting 512 channels back to 256 with ReLU and spatial dropout; the final classifier and the three auxiliary heads are likewise 3×3 convolutions that produce two-channel logits.

Auxiliary prediction heads are attached to the three fused scales so that each interaction stage receives direct segmentation supervision. Let y denote the final logits and $\{y_l\}_{l=1}^{3}$ the auxiliary outputs. The training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{ce}}(y, Y_q) + \sum_{l=1}^{3} \lambda_l \mathcal{L}_{\mathrm{ce}}(y_l, Y_q), \tag{9}$$

where $\mathcal{L}_{\mathrm{ce}}$ is pixel-wise cross-entropy and λ_l balances the auxiliary terms. Deep supervision prevents the sparse foreground signal from being diluted during long-range decoding and encourages meaningful support-driven selection at every resolution.

During training, only LPA, the semantic projections, LSA, SSI, and decoder are optimized. At inference, support prototypes are computed once for an episode and reused to segment its query images. The complete design consequently preserves a frozen and general semantic basis, introduces only compact episode-conditioned adaptation, and spends local modeling capacity specifically on boundary recovery. These roles make the contributions of the individual modules complementary rather than redundant.

4. Experiments

4.1. Experimental Settings

Dataset. We evaluate S^{3}DNet on FSSD-12, a few-shot strip-steel surface defect segmentation dataset constructed by Feng et al. [6], integrating defective images from four public industrial benchmarks (GC10-DET [55], X-SDD [56], SD-900 [57], and Surface Defects-4i [5]) with pixel-level annotations; all defective images are uniformly cropped and resized to 200×200. The twelve defect categories are abrasion-mask, iron-sheet-ash, liquid, oxide-scale, oil-spot, water-spot, patch, punching, red-iron sheet, scratch, roll-printing, and inclusion, and the number of defective images in each category is 50 to avoid a long-tailed class distribution. The dataset includes common but visually diverse industrial patterns such as elongated scratches, roll-printing defects, inclusions, patches, and abrasion-like regions. Their scale and foreground ratio vary considerably, and several categories contain weak contrast or highly textured backgrounds. These properties make the benchmark suitable for examining both semantic transfer and fine localization.

FSSD-12 follows a three-fold class split, as detailed in Table 1. In every fold, four defect categories are reserved for testing and the remaining eight categories are used for training, and the class sets of different folds do not overlap. Therefore, the model cannot solve the task by memorizing the test-category appearance. Instead, it must use the support set to define the target of each episode. We retain this official class-level cross-validation protocol and report Fold-0, Fold-1, Fold-2, and the arithmetic mean.

Table 1. Statistics and fold assignment of FSSD-12. Each category contains at most 50 defective images of 200 × 200 pixels with pixel-level masks; the test classes of different folds do not overlap, and the remaining eight categories of each fold are used for training.

Test Fold	Test Defect Classes	Images per Class
Fold-0	abrasion-mask, iron-sheet-ash, liquid, oxide-scale	50
Fold-1	oil-spot, water-spot, patch, punching	50
Fold-2	red-iron sheet, scratch, roll-printing, inclusion	50

Episodic protocol. Training and testing use episodic sampling. Each episode selects one target category, constructs a K-shot support set with binary masks, and samples one query image from the same category. We evaluate $K = 1$ and $K = 5$. At test time, every defective image of each unseen test class serves as the query of exactly one episode, and the K-shot support set is randomly sampled from the remaining defective images of the same class, so the episodic randomness at test time stems from the support draw rather than from the query enumeration. Support images are drawn uniformly at random without replacement, so the selection involves no manual curation; since every test image is queried exactly once, the evaluation probes the support distribution of each class through repeated random draws rather than relying on a single hand-picked support set. One fold therefore contains one episode per test image, i.e., at most 200 episodes under the 50-image-per-class cap of the benchmark, and the class-wise and fold-wise scores are averaged over these episodes. The residual variability across support draws is quantified by the four-trial evaluation in Section 4.2, where the fold-level standard deviation never exceeds 0.22 points. Finally, the three-fold mean is reported. All compared configurations use the same input processing, loss, and episode protocol so that the effect of the proposed modules can be isolated.

The original annotated image resolution is 200 × 200. Each image and mask is first resized to 224 × 224 for network processing. Before DINOv3, the image is bilinearly enlarged to 448 × 448, yielding a 28 × 28 token grid. The prediction is resized back to 200 × 200 before evaluation. We used nearest-neighbor interpolation for binary support masks to avoid introducing fractional labels.

Evaluation metrics. We adopt mean intersection over union (mIoU) and foreground–background IoU (FB-IoU). For category c, the intersection over union is

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \tag{10}$$

where TP, FP, and FN denote true-positive, false-positive, and false-negative pixels. mIoU averages the category-wise IoU over the test classes and reflects balanced segmentation quality across defect types. FB-IoU treats foreground and background as two semantic regions and averages their IoU. It is informative for industrial images because the background often occupies most pixels; reporting it together with mIoU prevents a foreground improvement from being interpreted without considering background separation.

Implementation details. We implemented S^3DNet in PyTorch 2.10 on Ubuntu 20.04 and conducted the experiments on an NVIDIA GeForce RTX 3090 with 24 GB memory and an Intel Xeon Platinum 8336C CPU at 2.30 GHz. DINOv3 ViT-S/16 and ViT-B/16 initialize the Small and Base semantic encoders, respectively. All DINOv3 parameters are frozen, including the patch-embedding layer, the twelve Transformer blocks, and the final layer normalization. No LayerNorm adaptation or other within-backbone modification is applied, and no exponential moving average of the weights is used for stabilization; training stability is instead supported by the frozen backbone itself and the zero-initialized LPA update. Twelve intermediate-layer outputs are obtained, and layers 5, 8, and 11 are

selected. LPA uses rank $r = 8$, the semantic projection dimension is 256, and the LSA channels are 16, 32, and 64. Fusion blocks use spatial dropout with probability 0.2.

We optimized the network for 200 epochs with a batch size of 2, using stochastic gradient descent with an initial learning rate of 10^{-3}, momentum 0.9, and weight decay 10^{-4}. The final classifier and the three auxiliary heads are supervised with equal weights. Only the adapters, projections, spatial branch, interaction modules, and decoder are updated. The same augmentation and optimization policy is retained for all ablation variants. This setting allows the semantic foundation to remain stable and restricts learning to the support-conditioning and spatial-refinement components.

4.2. Comparison with Representative Methods

To evaluate the effectiveness of S^3DNet, we compare it with representative prototype-, prior-, correlation-, graph-, and disentanglement-based few-shot segmentation models. Anomaly-detection methods such as AnomalyDINO are deliberately excluded from this comparison. They assume normal-only reference data, score deviations from learned or stored normal appearance, and produce threshold-based anomaly maps that are evaluated with AUROC-type metrics, whereas episodic FSS predicts a class-specific mask for the category specified by an annotated support set and is evaluated with per-class mIoU; combining the two protocols in one table would therefore be an unfair numerical comparison. The VGG-16 group contains PANet, PFENet, PMMs, CPANet, and DRNet. The ResNet-50 group contains CANet, PGNet, PMMs, PFENet, SCL, HSNet, TGRNet, DCP, CPANet, SCCAN, and DRNet. We retain the public results under the original backbone of every method. The proposed Small and Base models are listed in a separate DINOv3 group. This organization follows the comparison style of the source benchmark and avoids presenting a cross-backbone table as if it were a controlled backbone-only experiment.

Overall quantitative comparison. As shown in Table 2, both S^3DNet configurations achieve strong results in the 1-shot and 5-shot settings. DINOv3-Base obtains 69.2% mean mIoU and 80.1% FB-IoU with one support image. With five support images, the scores increase to 69.8% and 80.5%. Among the listed CNN methods, CPANet provides the strongest mean mIoU, reaching 61.5% in 1-shot and 62.6% in 5-shot. Relative to these values, S^3DNet-B improves mIoU by 7.7 and 7.2 percentage points, respectively. The simultaneous improvement in FB-IoU indicates that the gain is not obtained by indiscriminately enlarging the foreground; the separation between defect and background also becomes more reliable.

The Small configuration reaches 68.1%/68.6% mean mIoU and 79.5%/79.8% FB-IoU. Its performance is already substantially above the CNN group, suggesting that the proposed architecture does not depend on the largest encoder. The gain can be attributed to two complementary factors. DINOv3 provides a transferable semantic representation for unseen categories, while the support-driven adapters and spatial path convert that representation into a defect-specific dense prediction. A frozen foundation feature alone would not provide this episode-specific conditioning, and a lightweight spatial branch alone would lack robust semantics.

Fold-wise analysis. Fold-2 is the most difficult split for nearly all methods. For example, the 1-shot mIoU of the strongest ResNet-50 CPANet decreases from 66.0% on Fold-0 and 64.0% on Fold-1 to 54.6% on Fold-2. S^3DNet-S obtains 61.3% and S^3DNet-B obtains 61.5% on the same fold. The remaining gap between Fold-2 and the other folds shows that certain held-out defect categories still present severe appearance ambiguity. Even so, the proposed model reduces the degradation through its semantic foundation and multiscale spatial refinement.

Table 2. Performance comparison on FSSD-12 (%). The CNN results retain their originally reported backbones. Bold values indicate the better of the two S^{3}DNet configurations.

Backbone	Method	mIoU (1-Shot)				FB-IoU	mIoU (5-Shot)				FB-IoU
		Fold-0	Fold-1	Fold-2	Mean		Fold-0	Fold-1	Fold-2	Mean	
VGG-16	PANet [25]	54.7	45.7	40.7	47.0	69.1	54.7	50.7	44.8	50.1	71.9
	PFENet [30]	50.9	62.3	43.4	52.5	65.9	54.1	67.3	46.3	55.9	68.7
	PMMs [28]	46.3	54.2	42.5	47.7	63.9	46.4	54.5	43.6	48.2	64.2
	CPANet [6]	50.8	54.6	50.7	52.0	69.1	53.3	54.3	52.4	55.4	69.7
	DRNet [38]	46.6	37.6	40.1	41.4	61.5	49.9	50.0	48.0	46.6	62.4
ResNet-50	CANet [26]	54.4	54.2	48.2	52.3	67.8	56.1	56.0	51.2	54.5	69.2
	PGNet [35]	57.9	46.4	52.3	52.2	67.5	49.3	54.2	53.9	52.5	70.1
	PMMs [28]	56.7	52.7	40.9	50.1	66.7	56.9	52.9	41.2	50.4	67.2
	PFENet [30]	58.9	55.3	51.6	55.3	70.3	59.9	56.0	52.2	56.0	74.0
	SCL [34]	57.5	50.2	46.3	51.3	68.4	57.9	50.4	47.4	51.9	68.8
	HSNet [32]	50.2	53.0	41.8	48.4	67.9	56.0	60.9	47.3	54.7	71.6
	TGRNet [5]	61.8	62.0	48.3	57.7	73.6	62.4	59.7	47.8	58.5	75.1
	DCP [36]	58.3	51.7	43.1	51.0	70.6	55.9	47.0	55.7	52.9	70.9
	CPANet [6]	66.0	64.0	54.6	61.5	76.1	66.5	64.9	56.3	62.6	76.3
	SCCAN [37]	53.7	60.7	50.9	55.1	70.7	56.9	61.1	53.4	57.1	72.2
	DRNet [38]	64.5	63.8	44.8	57.7	72.7	66.6	65.9	54.6	62.4	76.6
DINOv3	S^{3}DNet (Small)	**70.3**	72.7	61.3	68.1	79.5	**70.7**	73.6	61.6	68.6	79.8
	S^{3}DNet (Base)	69.2	**77.0**	**61.5**	**69.2**	**80.1**	70.6	**77.1**	**61.7**	**69.8**	**80.5**

The effect of backbone size is not uniform across folds. Base is especially beneficial on Fold-1, improving over Small by 4.3 points in 1-shot and 3.5 points in 5-shot. On Fold-0, Small is 1.1 points higher in 1-shot and the two models are nearly identical in 5-shot. Fold-2 also differs by only 0.2/0.1 points. This asymmetry is explained by the characteristics of the held-out categories. The Fold-1 test classes (oil-spot, water-spot, patch, and punching) are amorphous, low-contrast stains and imprints with weak shape priors, so separating them depends mainly on global appearance and subtle channel-wise semantics, and the richer frozen representation of ViT-B yields the largest benefit. The Fold-2 classes (scratch, roll-printing, inclusion, and red-iron sheet) are dominated by thin, high-frequency structures whose accuracy is bounded by the patch-grid resolution rather than by semantic capacity, so both encoders saturate at the same level. On Fold-0, whose classes are broad, texture-like regions already covered by ViT-S semantics, the smaller adaptation head is equally effective, and the residual one-point advantage of Small is consistent with a mild regularization effect of its fewer trainable parameters under scarce supervision. Statistically, the seed-paired per-fold differences confirm this reading: the 95% confidence intervals of the Base-minus-Small mIoU difference over the four episode seeds are $[4.12, 5.03]$ in favor of Base on Fold-1 and $[-1.58, -0.92]$ in favor of Small on Fold-0, whereas on Fold-2 the interval $[-0.21, 0.26]$ contains zero; the exact Wilcoxon signed-rank test yields $p = 0.125$ on Folds 0 and 1—the smallest two-sided value attainable with four runs—and $p = 1.00$ on Fold-2, so the asymmetry between folds is systematic rather than accidental. A Friedman test over the fold-wise mean mIoU of the thirteen methods in the ResNet-50 and DINOv3 groups of Table 2 likewise rejects the null hypothesis of equal performance ($\chi^2(12) = 27.3$, $p = 0.014$). These observations suggest that a larger semantic encoder is most useful when the held-out categories demand stronger representation capacity, while the lightweight Small encoder can be equally effective when the support-conditioned spatial cues are already discriminative.

Effect of support-shot number. Increasing the support set from one to five images improves the mean mIoU of Small from 68.1% to 68.6% and Base from 69.2% to 69.8%. FB-IoU rises by 0.3 and 0.4 points, respectively. The improvement is consistent but moderate.

This result indicates that LPA and SSI can extract a useful target direction from a single annotated example. Additional support images mainly stabilize the prototype, compensate for within-class appearance variation, and refine the boundary of weak regions. The result is desirable in practice because, for instance, rare industrial defects may not provide five high-quality annotations.

Evaluation stability. Because the support sets are drawn at random, we repeated the complete evaluation of both configurations over four independent random support-sampling trials and report the average performance with standard deviations over these trials. The results are highly stable: the mean mIoU is 68.10 ± 0.08 in 1-shot and 68.55 ± 0.17 in 5-shot for S^3DNet-S, and 69.18 ± 0.05 in 1-shot and 69.63 ± 0.13 in 5-shot for S^3DNet-B, while the corresponding FB-IoU standard deviations remain below 0.10 points. The 95% confidence intervals of the mean mIoU are $[67.97, 68.23]$ and $[68.27, 68.83]$ for S^3DNet-S in the 1-shot and 5-shot settings, and $[69.10, 69.26]$ and $[69.43, 69.83]$ for S^3DNet-B, all of which lie far above the best CNN result (61.5% and 62.6% mean mIoU). Across all folds and runs, the fold-wise standard deviation never exceeds 0.22 points. Since the improvement over the strongest CNN baseline (7.7 and 7.2 points of mean mIoU in the 1-shot and 5-shot settings) is more than an order of magnitude larger than this episodic sampling noise, the reported gains do not depend on a favorable episode draw.

4.3. Computational Performance

In addition to segmentation accuracy, we compare parameter count, floating-point operations, peak GPU memory, GPU execution time, and end-to-end inference time. The parameter and FLOP values include the frozen DINOv3 encoder because it remains part of the deployed model. GPU time measures the principal GPU computation, whereas E2E time also reflects data movement and other framework overhead. Reporting both values is important because parameter count and FLOPs do not fully determine wall-clock latency.

As summarized in Table 3, S^3DNet-S provides the more deployment-friendly operating point. It contains 26.3 M parameters, which is smaller than every listed competitor, and requires 128.1 GFLOPs. Its peak GPU memory is 2214 MB, GPU time is 35.5 ms, and end-to-end time is 40.0 ms. The number of FLOPs is higher than those of PFENet, TGRNet, DCP, and CPANet because DINOv3 processes a 448×448 input to produce a denser token grid. Nevertheless, its latency remains moderate, illustrating that modern Transformer operations can execute efficiently on the GPU.

Table 3. Computational performance comparison.

Method	Params.	FLOPs	GPU Mem.	GPU Time	E2E Time
PFENet [30]	34.5 M	85.68 G	1126 MB	26.32 ms	31.33 ms
TGRNet [5]	31.2 M	73.57 G	1280 MB	80.29 ms	115.42 ms
DCP [36]	35.0 M	71.09 G	1331 MB	42.16 ms	43.59 ms
CPANet [6]	30.3 M	85.20 G	1741 MB	29.48 ms	33.42 ms
SCCAN [37]	35.0 M	240.44 G	2356 MB	72.40 ms	75.69 ms
DRNet [38]	330.8 M	151.49 G	3954 MB	132.96 ms	135.28 ms
S^3DNet-B	90.7 M	331.1 G	4092 MB	41.2 ms	45.8 ms
S^3DNet-S	26.3 M	128.1 G	2214 MB	35.5 ms	40.0 ms

S^3DNet-B increases the parameter count to 90.7 M and the number of FLOPs to 331.1 G. The corresponding peak memory is 4092 MB and the end-to-end time is 45.8 ms. In return, it improves mean mIoU over Small by 1.1 points in 1-shot and 1.2 points in 5-shot. Thus, the two configurations serve different requirements. Small is suitable for real-time or memory-constrained deployment, whereas Base is preferable when segmentation accuracy has priority and sufficient GPU resources are available. The fact that Base has substantially

more FLOPs but only 5.8 ms additional end-to-end time also confirms that FLOPs, memory access, kernel fusion, and parallel utilization jointly determine practical speed.

4.4. Ablation Studies

Ablation setting. We use DINOv3-Small as a fixed frozen encoder and construct a baseline that retains the semantic projection and basic decoder but removes LPA, LSA, and SSI. We then introduce LPA or LSA individually, combine LPA and LSA, and finally add SSI to form the complete model. Every variant uses the same three-fold protocol, training schedule, and loss. The results in Table 4 therefore measure the cumulative effect of support-conditioned adaptation, spatial compensation, and their explicit interaction.

Table 4. Component ablation on FSSD-12 with DINOv3-Small (%).

Setting	LPA	LSA	SSI	mIoU (1-Shot)				FB-IoU	mIoU (5-Shot)				FB-IoU
				Fold-0	Fold-1	Fold-2	Mean		Fold-0	Fold-1	Fold-2	Mean	
Baseline	–	–	–	61.2	70.7	56.6	62.8	77.0	61.4	70.8	56.8	63.0	77.1
LPA only	✓	–	–	62.0	70.9	57.1	63.3	77.4	62.5	71.1	57.3	63.6	77.5
LSA only	–	✓	–	65.1	71.1	57.7	64.6	78.1	65.2	71.0	57.6	64.6	78.2
LPA + LSA	✓	✓	–	68.7	71.6	59.0	66.4	78.8	69.2	72.1	59.4	66.9	79.0
Full model	✓	✓	✓	70.3	72.7	61.3	68.1	79.5	70.7	73.6	61.6	68.6	79.8

Effectiveness of LPA. Adding only LPA increases the mean mIoU from 62.8% to 63.3% in 1-shot and from 63.0% to 63.6% in 5-shot. FB-IoU increases by 0.4 points in both settings. Although the gain is smaller than that of the spatial branch, it is consistent on all three folds. This verifies that the generic frozen token representation benefits from an episode-specific semantic direction. LPA is also deliberately compact; its role is not to relearn the encoder, but to correct the channel emphasis according to the masked support foreground.

The improvement becomes slightly larger in 5-shot, where multiple supports provide a more stable prototype. This trend agrees with the design of LPA: averaging support directions reduces the effect of a single atypical sample, and the low-rank mapping can produce a cleaner query update. The remaining moderate magnitude is reasonable because the baseline already benefits from strong DINOv3 semantics. LPA refines those semantics rather than replacing them.

Effectiveness of LSA. Using LSA alone yields 64.6% mean mIoU in both 1-shot and 5-shot, improving the baseline by 1.8 and 1.6 points. FB-IoU reaches 78.1% and 78.2%. The larger isolated gain confirms that spatial resolution and local structure are critical bottlenecks in strip-steel segmentation. Even a strong semantic representation may miss a narrow line or produce a smooth boundary; the three shallow convolutional stages expose exactly this missing information.

LSA improves Fold-0 by 3.9/3.8 points and Fold-2 by 1.1/0.8 points, whereas the change on Fold-1 is smaller. This variation indicates that the held-out categories have different spatial characteristics. Categories dominated by narrow or fragmented geometry benefit most strongly from explicit local features, while categories with distinctive global appearance can already be handled well by the semantic baseline.

Complementarity of LPA and LSA. Combining LPA and LSA, even before introducing SSI, raises the mean mIoU to 66.4%/66.9%. These values are 3.6/3.9 points above the baseline and exceed either individual module. The joint gain is not simply redundant: LPA improves the class direction, whereas LSA restores spatial detail. A more accurate semantic feature makes the spatial information easier to associate with the target, and a sharper spatial feature allows the semantic decision to be expressed as a precise mask.

Effectiveness of SSI. Adding SSI to the combined model produces the final 68.1%/68.6% mean mIoU and 79.5%/79.8% FB-IoU. Relative to the ungated LPA+LSA combination, SSI contributes another 1.7 points in both shot settings. Relative to the baseline, the complete model improves mIoU by 5.3/5.6 points and FB-IoU by 2.5/2.7 points. The gain verifies that merely providing two feature paths is insufficient. Their interaction must be conditioned on the support category so that background texture from LSA is suppressed before fusion.

The improvement is again consistent across all folds. On Fold-2, SSI increases 1-shot mIoU from 59.0% to 61.3% and 5-shot mIoU from 59.4% to 61.6%, showing particular value on the most difficult split. This supports the proposed interpretation: when semantic ambiguity and texture interference are both strong, a learned support-conditioned spatial gate provides a more reliable connection than unconditional concatenation.

To isolate the contribution of the fusion strategy itself, Table 5 compares the full model with two ablated variants in which only one of the two SSI mechanisms is active: applying the spatial gate without the concatenation fusion, or concatenating the LSA features with the semantic maps without gating. Either mechanism alone improves the LPA + LSA baseline only moderately (66.8 and 66.6 vs. 66.4 in Table 4), whereas combining them reaches 68.1, exceeding either single mechanism by 1.3–1.5 points on average and achieving the highest FB-IoU (79.5). The two mechanisms are therefore complementary rather than redundant: the gate selects class-discriminative locations, and the concatenation fusion allows the decoder to exploit the filtered detail; the lower FB-IoU of both variants shows that removing either one leaves the majority background to dominate the prediction.

Table 5. Ablation of the SSI fusion strategy (1-shot, %).

Fusion Strategy	Fold-0	Fold-1	Fold-2	Mean	FB-IoU
Full model (gate + concatenation)	70.3	72.7	61.3	68.1	79.5
Gate only	68.6	70.9	60.9	66.8	77.7
Concatenation only	66.0	72.3	61.5	66.6	78.7

Effect of DINOv3 input resolution. FSSD-12 images are natively 200 × 200, and strip-steel defects such as scratches and roll prints occupy only a small fraction of this area. Because DINOv3 tokenizes the image at a fixed 1/16 stride, the input resolution directly determines the density of the semantic grid: a 224 × 224 input yields only 14 × 14 tokens, so a thin defect of a few pixels falls inside a single patch and its evidence is mixed with background before matching begins. Table 6 quantifies this effect with all other settings unchanged. Enlarging the input to 448 × 448 (a 28 × 28 grid) raises the mean mIoU from 59.8 to 69.2 in 1-shot and from 59.1 to 69.8 in 5-shot, with consistent gains on all three folds. The intermediate 336 × 336 input (21 × 21 tokens) brings no comparable gain, which matches the geometric analysis: at that scale most thin defects are still confined to one or two tokens, and the 21 × 21 grid no longer aligns with the coarse scale of the LSA pyramid, forcing irregular upsampling of the semantic maps, whereas the 28 × 28 grid coincides exactly with the coarse LSA scale and assigns multiple tokens even to narrow defects. All other experiments therefore adopt the 448 × 448 input.

Table 6. Ablation of the DINOv3 input resolution with DINOv3-Base (%).

Input	Token Grid	mIoU (1-Shot)				mIoU (5-Shot)			
		Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean
224 × 224	14 × 14	61.7	68.7	49.2	59.8	60.2	67.1	49.9	59.1
336 × 336	21 × 21	61.2	69.3	51.9	60.8	61.1	69.1	52.7	61.0
448 × 448	28 × 28	69.2	77.0	61.5	69.2	70.6	77.1	61.7	69.8

Effect of the segmentation loss. Although defect segmentation is class-imbalanced, the proposed model suppresses background responses architecturally, through the support-conditioned gating of SSI and the frozen DINOv3 semantics. So the segmentation head operates on less imbalanced evidence than a from-scratch detector. Table 7 therefore compares the cross-entropy objective with two region-overlap alternatives on the Base configuration. Cross-entropy outperforms the Dice and combined CE + Dice losses by 5.3–7.2 points of mean mIoU, and the advantage is consistent across all three folds. This result is explained by the properties of region-overlap objectives: Dice optimizes global overlap with nearly uniform gradient weight, which dilutes the per-pixel boundary evidence on which thin scratches and roll prints depend, and the combined CE + Dice variant inherits the same dilution while further reducing the contribution of the pixel-wise cross-entropy term. Cross-entropy with equal per-pixel weighting thus preserves exactly the boundary-precision signal that the progressive decoder needs, and it is retained for all experiments.

Table 7. Ablation of the segmentation loss with DINOv3-Base (1-shot, %).

Loss	Fold-0	Fold-1	Fold-2	Mean	FB-IoU
Cross-entropy	69.2	77.0	61.5	69.2	80.1
Dice	70.3	65.5	55.9	63.9	77.7
CE + Dice	66.6	65.0	54.5	62.0	76.3

Effect of the pretrained backbone. To separate the contribution of pretraining from that of the framework itself, we replace the DINOv3 encoder with an ImageNet-pretrained ResNet-50, kept frozen under the same protocol, and evaluate the model with and without the innovation modules. As shown in Table 8, the framework remains beneficial on both backbones, improving the ResNet-50 variant from 54.0 to 55.0 in 1-shot and from 54.1 to 55.4 in 5-shot, and DINOv3-S from 62.8 to 68.1 and from 63.0 to 68.6; the gain, however, is far larger on the self-supervised encoder. Indeed, replacing DINOv3-S with ResNet-50 under the identical framework reduces the mean mIoU by 13.1 points, and the ResNet-50 variant (55.0) remains below the end-to-end trained CPANet (61.5). These results indicate that the advantage over the from-scratch CNN methods originates primarily from the billion-scale self-supervised representation, in line with the design premise of S^3DNet as a parameter-efficient transfer framework, while the proposed modules contribute an additional, backbone-independent margin through episode-conditioned adaptation and gated spatial recovery.

Table 8. Effect of the pretrained backbone (mean mIoU, %).

Backbone	Pre-Training	LPA/LSA/SSI	1-Shot	5-Shot
ResNet-50	ImageNet	–	54.0	54.1
ResNet-50	ImageNet	✓	55.0	55.4
DINOv3-S	Self-supervised	–	62.8	63.0
DINOv3-S	Self-supervised	✓	68.1	68.6
DINOv3-B	Self-supervised	✓	69.2	69.8

4.5. Cross-Dataset Generalization

To examine whether the frozen foundation representation generalizes beyond the training benchmark, we evaluate S^3DNet directly on Surface Defects-4i [5] with the weights trained on FSSD-12, without any fine-tuning on the target dataset. Surface Defects-4i covers metal generic surface defects from different materials and imaging conditions than FSSD-12, and we apply the same episodic protocol under its three-fold class split. Table 9 reports

the results. Without observing any target-domain image, the model achieves 43.28/42.29 mean mIoU and 59.47/60.67 FB-IoU in the 1-shot and 5-shot settings. The drop relative to the in-domain results (69.2/69.8 mean mIoU) is expected, since the LPA update, the fusion weights, and the decoder are trained on the defect style of FSSD-12, whereas Surface Defects-4i exhibits different materials, textures, and defect morphologies. Nevertheless, the frozen encoder still supplies transferable semantics, and the support-conditioned pipeline segments unseen categories of a different dataset with reasonable accuracy, which supports the domain generality of the frozen representation. Interestingly, the 5-shot setting does not further improve over 1-shot under this shift, suggesting that additional in-domain supports do not necessarily align better with the changed target appearance. Narrowing this gap, for example, by adapting the prototypes or the low-rank update to a few target-domain samples, is left for future work.

Table 9. Cross-dataset evaluation: training on FSSD-12 and testing on Surface Defects-4i without fine-tuning (%).

Setting	mIoU (1-Shot)				FB-IoU	mIoU (5-Shot)				FB-IoU
	Fold-0	Fold-1	Fold-2	Mean		Fold-0	Fold-1	Fold-2	Mean	
S^3DNet (FSSD-12 → SD-4i)	40.61	50.32	38.90	43.28	59.47	38.57	50.82	37.48	42.29	60.67

4.6. Qualitative Analysis

To visually compare localization and boundary quality, Figure 5 presents the support image and mask, the query image and ground truth, and predictions from PFENet, TGRNet, CPANet, and S^3DNet. The examples cover diverse defect appearances, including thin structures, low-contrast regions, irregular patches, and textured backgrounds. All methods receive the same support information.

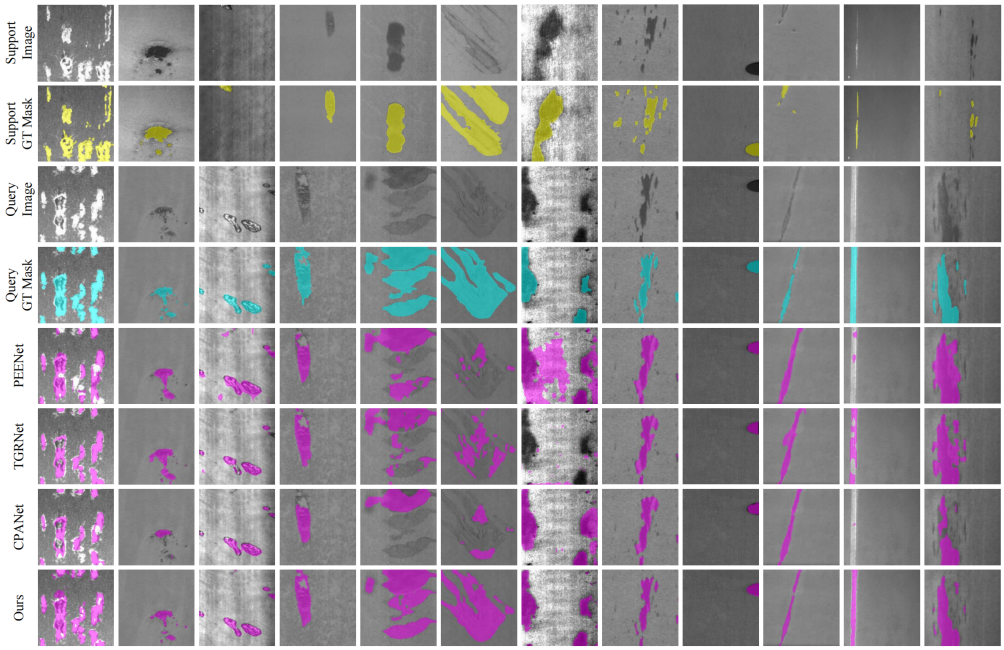


Figure 5. Qualitative comparison on FSSD-12. Each group contains the support image and mask, query image and ground truth, followed by predictions of representative competitors and S^3DNet.

PFENet tends to generate coarse or fragmented foreground responses. When the defect contrast is weak, part of the target disappears; when the background texture is strong, isolated responses remain. TGRNet covers the main defect more completely in several ex-

amples, but its predictions can expand beyond the ground-truth boundary or contain local holes. CPANet provides a stronger overall baseline and recovers the principal foreground in many cases, yet narrow structures and weak edges are still partially under-segmented.

In contrast, S^{3}DNet maintains better connectivity for elongated defects and produces a more complete mask for low-contrast regions. The prediction boundary is also more closely aligned with the ground truth, and scattered activation in the surrounding texture is reduced. The visual improvement is consistent with the quantitative changes in both mIoU and FB-IoU. LPA helps concentrate the semantic response on the support-defined category, while LSA and the fine-scale SSI branch recover the local contour. The examples also illustrate why the two paths are complementary: semantics suppress visually similar background, and spatial detail prevents the semantic response from becoming overly smooth.

4.7. Failure Cases and Discussion

Although S^{3}DNet performs well on most samples, it does not eliminate all industrial ambiguities. Figure 6 summarizes representative errors under complex background interference, extremely thin structures, and low contrast.

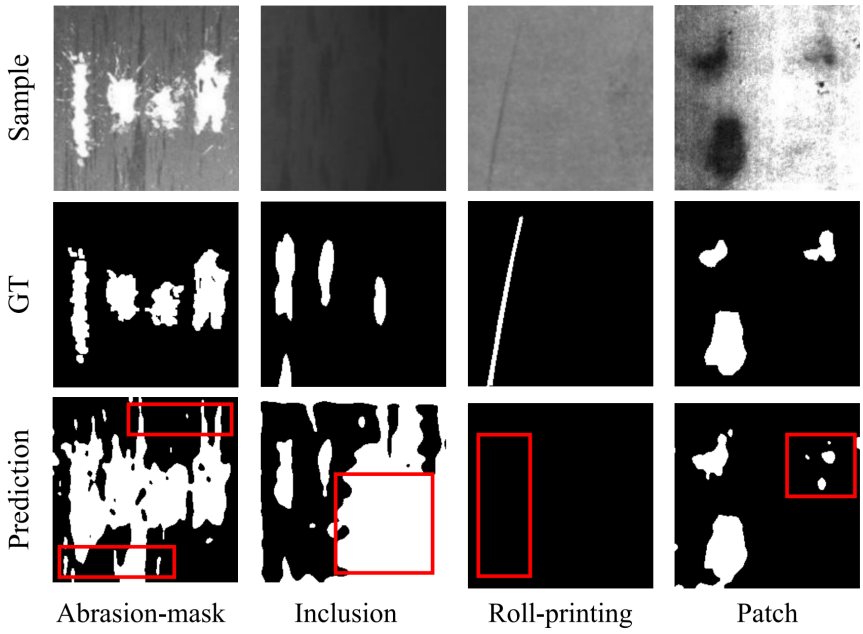


Figure 6. Failure cases. Red boxes highlight false-positive expansion, missed thin structures, and scattered responses under strong texture or low contrast.

Texture-induced false positives. In the abrasion-mask and inclusion examples, the prediction expands into background regions and can form a large false-positive area. Strong local texture may be close to the support prototype in feature space. Once the resulting semantic similarity is overestimated, the spatial gate repeatedly preserves that region at several scales. The top-down decoder can then propagate the error. This behavior is related to the sensitivity of prototype-based priors to noisy context [31]. Existing research offers concrete remedies for this failure mode. A background-contrastive matching rule, in the spirit of the triplet reasoning of TGRNet [5], would compare each query location with both a foreground and a background prototype and retain spatial evidence only when the foreground margin is positive, so that texture equally similar to both prototypes is rejected. Robust prototype estimation is a second remedy: prototype mixtures [28] and selective prototype construction [22] suppress ambiguous support regions before matching, and the noise-filtered prior of PFENet++ [31] performs the analogous correction on the

query side. In addition, consistency-based guidance [34] indicates that enforcing agreement between interacting branches or adjacent scales suppresses unreliable responses before fusion. Among these options, the background-contrastive gate is the most compatible with our decoupled design because it reuses the prototype machinery of SSI without introducing additional parameters; we will evaluate this extension systematically in future work.

Extremely thin structures. In the roll-printing example, an elongated low-contrast line is partially missed. The defect occupies very few patch positions even after input upsampling, and its evidence is dominated by high-frequency boundaries. The semantic token at a local position may therefore mix foreground and background. Although LSA provides a higher-resolution spatial feature, its gate is still derived from the token-based semantic map. This creates a resolution bottleneck that is consistent with known limitations of Transformer representations for local texture [52,54].

Scattered uncertain responses. The patch example contains several small false-positive components. When defect and background texture are highly similar, the support-driven similarity map may fluctuate locally. The sigmoid gate is continuous rather than binary, so weak erroneous responses can survive the fusion. Spatial post-processing could remove some isolated components, but a learned uncertainty or consistency mechanism would be preferable because it could distinguish valid small defects from noise.

These cases identify three current limitations: robustness of the support prototype under appearance noise, spatial resolution for structures thinner than the patch grid, and calibration of similarity across feature levels. A promising extension is to enforce consistency among several DINOv3 layers and use the disagreement as an uncertainty estimate. High-resolution local tokens or frequency-aware refinement could further improve thin-defect recovery. In addition, although the cross-dataset evaluation on Surface Defects-4i (Section 4.5) indicates that the frozen representation transfers across benchmarks, a visible domain gap remains; broader cross-dataset studies under different sensors and production lines are needed to fully measure the domain generalization of the proposed design.

From a deployment perspective, the results indicate a clear operating trade-off. S^3DNet-S is compact and already provides strong accuracy, making it suitable as the default configuration. S^3DNet-B increases the semantic capacity and achieves the highest mean scores, but the advantage is concentrated on the more challenging fold and comes with higher memory and FLOPs. This flexibility is useful in industrial practice because the same spatial–semantic design can be paired with different frozen encoder sizes according to the hardware budget.

5. Conclusions

In this paper, we proposed S^3DNet, a support-driven spatial–semantic decoupling network for few-shot strip-steel surface defect segmentation. The model preserves a self-supervised DINOv3 encoder as a stable semantic foundation and learns only task-specific adaptation, spatial, interaction, and decoding components. LPA converts the masked support prototype into a low-rank update of the query tokens, enabling episode-specific semantic conditioning without full backbone fine-tuning. LSA independently restores multiscale texture and boundary cues, while SSI transforms support–query similarity into spatial gates so that local evidence is fused only at target-relevant positions. Experiments on FSSD-12 demonstrate consistent improvements under both 1-shot and 5-shot protocols. The Base configuration achieves the highest mean accuracy, whereas the Small configuration provides a favorable balance among parameters, memory, latency, and segmentation quality. Component ablations verify that semantic adaptation, spatial refinement, and gated interaction are mutually complementary. Qualitative and failure-case analyses further show

that the method improves elongated and low-contrast defects but remains challenged by highly repetitive texture and structures thinner than the patch grid.

Future work will investigate cross-layer similarity calibration, uncertainty-aware prototype estimation, and higher-resolution local refinement for more robust cross-domain industrial deployment.

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Abbreviations

The following abbreviations are used in this manuscript:

FSS	Few-shot semantic segmentation
LPA	Low-rank prototype adapter
LSA	Low-weight spatial adapter
SSI	Spatial–semantic interaction module
ViT	Vision Transformer
mIoU	Mean intersection over union
FB-IoU	Foreground–background intersection over union
CNN	Convolutional neural network
FLOPs	Floating-point operations
GPU	Graphics processing unit

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